README & Requirements

Project: Enterprise Customer Churn Prediction & Advanced Artificial Intelligence Analytics

Student: JATIN

README

Enterprise Customer Churn Prediction & Advanced Artificial Intelligence Analytics

Prepared by: JATIN

Institution: Government College, Dholpur

Programme: AICTE | IBM SkillsBuild Data Analytics with AI Internship 2026

Project overview

This project demonstrates an end-to-end customer churn analytics workflow: data quality checks, descriptive analysis, preprocessing, a Logistic Regression classification pipeline, evaluation metrics, and visual outputs.

Important data limitation

The included dataset is generated synthetic practice data. It is not real customer data and is not evidence about any real telecom company. Results are for demonstrating code and methodology only. If the internship instructions require a public or mentor-provided dataset, replace the data-generation function with that approved dataset, cite its source, rerun the script, and update the report.

Files

• `churn_prediction.py` — complete executable analysis and machine-learning pipeline.

• `requirements.txt` — Python dependencies.

• `Project_Report.docx` — internship report.

• `README.md` — project instructions and limitations.

Run locally

1. Install Python 3.10 or newer.

2. Open a terminal in this folder.

3. Install dependencies:

```bash

pip install -r requirements.txt

```

4. Run:

```bash

python churn_prediction.py

```

5. Check the generated `outputs/` folder for CSV summaries, evaluation metrics, and charts.

Run in Google Colab

Upload `churn_prediction.py` and `requirements.txt` to Colab, install the requirements with `!pip install -r requirements.txt`, then run `!python churn_prediction.py`.

Methodology

1. Generate a reproducible demonstration dataset using a fixed random seed.
2. Inspect dimensions, missing values, and duplicate rows.
3. Summarise churn rate, monthly charges, and tenure.
4. Split features and target into stratified training and test sets.
5. Impute/scale numeric fields and impute/one-hot encode categorical fields.
6. Train a class-balanced Logistic Regression pipeline.
7. Report accuracy, precision, recall, F1-score, ROC-AUC, and a confusion matrix.
8. Save plots and tables in `outputs/`.

Responsible interpretation

Model metrics depend on the synthetic patterns built into this demo and should not be used for business decisions. On a real approved dataset, review data permissions, missingness, class imbalance, leakage, fairness, and performance on a held-out test set.

Requirements

```
numpy>=1.24
pandas>=2.0
matplotlib>=3.7
scikit-learn>=1.3
```